

HelperLearner - Database Schema & Entity Relationship Diagram

1. Entity Relationship Diagram (ER Diagram)

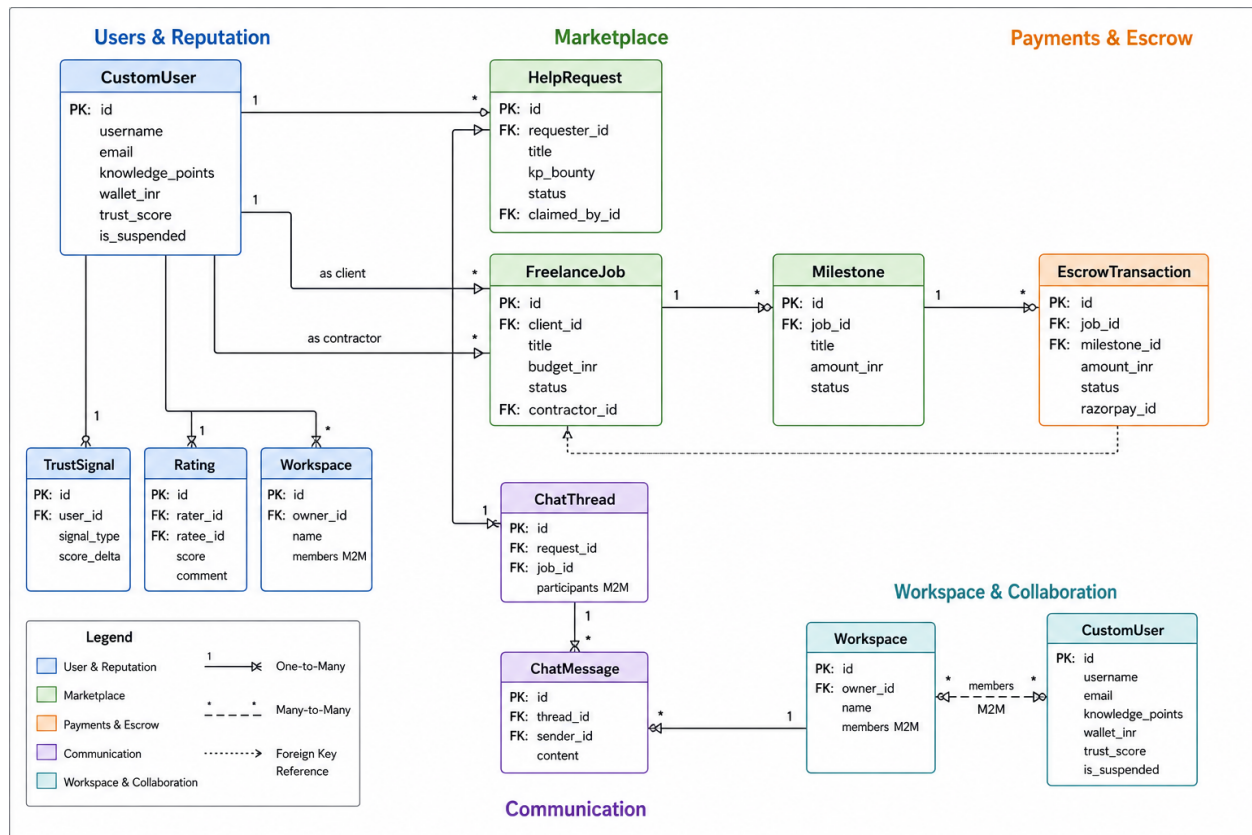


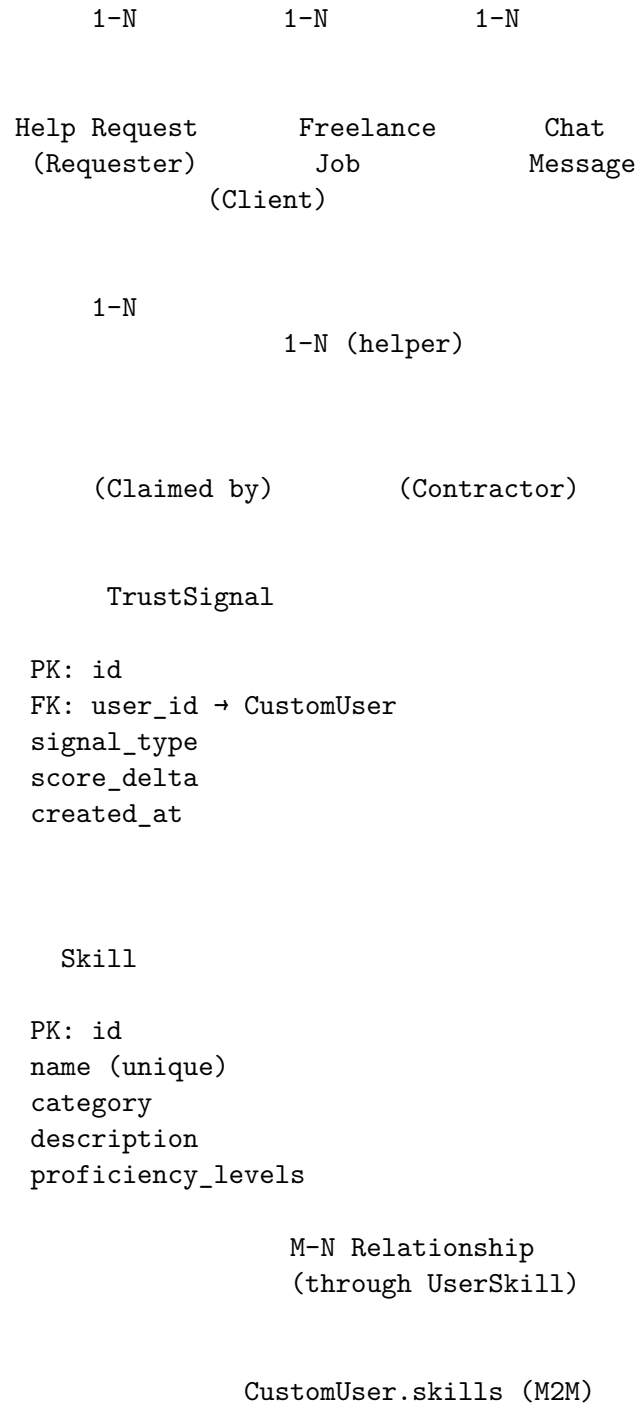
Figure 1: Entity Relationship Diagram — Complete database entity relationships with color-coded groups

CustomUser

(Django User + Extended)

```

PK: id
username (unique)
email (unique)
bio
knowledge_points
wallet_inr
trust_score
is_suspended
compliance_verified
created_at
updated_at
    
```



2. Table Schemas (SQL DDL)

2.1 Authentication & User Management

```

-- CustomUser (Extended Django User Model)
CREATE TABLE auth_user (
  id SERIAL PRIMARY KEY,

```

```

username VARCHAR(150) NOT NULL UNIQUE,
email VARCHAR(254) NOT NULL UNIQUE,
password VARCHAR(128) NOT NULL,
first_name VARCHAR(150),
last_name VARCHAR(150),
is_staff BOOLEAN DEFAULT FALSE,
is_active BOOLEAN DEFAULT TRUE,
is_superuser BOOLEAN DEFAULT FALSE,
last_login TIMESTAMP,
date_joined TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);

CREATE TABLE accounts_customuser (
    -- Inherits from auth_user
    user_ptr_id INTEGER PRIMARY KEY REFERENCES auth_user(id) ON DELETE CASCADE,
    bio TEXT,
    knowledge_points INTEGER DEFAULT 100 CHECK (knowledge_points >= 0),
    last_kp_claim TIMESTAMP,
    wallet_inr NUMERIC(12,2) DEFAULT 0.00 CHECK (wallet_inr >= 0),
    trust_score FLOAT DEFAULT 0.0,
    trust_score_updated_at TIMESTAMP,
    compliance_verified BOOLEAN DEFAULT FALSE,
    is_suspended BOOLEAN DEFAULT FALSE,
    suspended_until TIMESTAMP,
    suspension_reason VARCHAR(255),
    notification_preference VARCHAR(12) DEFAULT 'both',
    ui_density VARCHAR(12) DEFAULT 'comfortable',
    notify_requests BOOLEAN DEFAULT TRUE,
    notify_jobs BOOLEAN DEFAULT TRUE,
    notify_chat BOOLEAN DEFAULT TRUE,
    notify_kp BOOLEAN DEFAULT TRUE,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    updated_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    INDEX idx_knowledge_points (knowledge_points),
    INDEX idx_wallet_inr (wallet_inr),
    INDEX idx_trust_score (trust_score),
    INDEX idx_is_suspended (is_suspended)
);

-- Skill Model
CREATE TABLE marketplace_skill (
    id SERIAL PRIMARY KEY,
    name VARCHAR(100) NOT NULL UNIQUE,
    category VARCHAR(50),
    description TEXT,
    proficiency_levels VARCHAR(255),
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);

```

```

-- UserSkill (M2M through table)
CREATE TABLE accounts_customuser_skills (
    id SERIAL PRIMARY KEY,
    customuser_id INTEGER NOT NULL REFERENCES accounts_customuser(user_ptr_id),
    skill_id INTEGER NOT NULL REFERENCES marketplace_skill(id),
    UNIQUE(customuser_id, skill_id)
);

```

2.2 Help Request System

```

-- HelpRequest Model
CREATE TABLE marketplace_helprequest (
    id SERIAL PRIMARY KEY,
    requester_id INTEGER NOT NULL REFERENCES accounts_customuser(user_ptr_id),
    title VARCHAR(255) NOT NULL,
    description TEXT NOT NULL,
    kp_bounty INTEGER NOT NULL CHECK (kp_bounty BETWEEN 1 AND 1000),
    difficulty VARCHAR(20) DEFAULT 'medium',
    status VARCHAR(50) DEFAULT 'open',
    tags VARCHAR(255),
    ai_summary TEXT,
    search_vector tsvector, -- For full-text search
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    updated_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    claimed_by_id INTEGER REFERENCES accounts_customuser(user_ptr_id),
    claimed_at TIMESTAMP,
    resolved_at TIMESTAMP,
    expires_at TIMESTAMP,

    -- Indexes
    INDEX idx_status (status),
    INDEX idx_created_at (created_at),
    INDEX idx_claimed_by (claimed_by_id),
    FULLTEXT INDEX idx_search_vector (search_vector)
);

```

```

-- Solution Model
CREATE TABLE marketplace_solution (
    id SERIAL PRIMARY KEY,
    request_id INTEGER NOT NULL REFERENCES marketplace_helprequest(id),
    helper_id INTEGER NOT NULL REFERENCES accounts_customuser(user_ptr_id),
    solution_text TEXT NOT NULL,
    code_snippet TEXT,
    is_approved BOOLEAN DEFAULT FALSE,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    updated_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    approved_at TIMESTAMP,

```

```

        approval_rating INTEGER, -- 1-5 stars
        approval_comment TEXT
    );

-- Rating Model
CREATE TABLE marketplace_rating (
    id SERIAL PRIMARY KEY,
    rater_id INTEGER NOT NULL REFERENCES accounts_customuser(user_ptr_id),
    ratee_id INTEGER NOT NULL REFERENCES accounts_customuser(user_ptr_id),
    score INTEGER NOT NULL CHECK (score BETWEEN 1 AND 5),
    comment TEXT,
    content_type VARCHAR(50), -- 'help_request' or 'freelance_job'
    object_id INTEGER NOT NULL,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,

    UNIQUE(rater_id, ratee_id, object_id)
);

-- Comment Model
CREATE TABLE marketplace_comment (
    id SERIAL PRIMARY KEY,
    author_id INTEGER NOT NULL REFERENCES accounts_customuser(user_ptr_id),
    request_id INTEGER REFERENCES marketplace_helprequest(id),
    content TEXT NOT NULL,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    updated_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    is_deleted BOOLEAN DEFAULT FALSE
);

```

2.3 Freelance Job System

```

-- FreelanceJob Model
CREATE TABLE marketplace_freelancejob (
    id SERIAL PRIMARY KEY,
    client_id INTEGER NOT NULL REFERENCES accounts_customuser(user_ptr_id),
    title VARCHAR(255) NOT NULL,
    description TEXT NOT NULL,
    budget_inr NUMERIC(12,2) NOT NULL CHECK (budget_inr > 0),
    status VARCHAR(50) DEFAULT 'open',
    deadline TIMESTAMP NOT NULL,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    updated_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    contractor_id INTEGER REFERENCES accounts_customuser(user_ptr_id),
    contract_status VARCHAR(50),

    INDEX idx_status (status),
    INDEX idx_deadline (deadline),
    INDEX idx_client_id (client_id)
);

```

```

);

-- Proposal Model (Contractors bid on jobs)
CREATE TABLE marketplace_proposal (
    id SERIAL PRIMARY KEY,
    job_id INTEGER NOT NULL REFERENCES marketplace_freelancejob(id),
    contractor_id INTEGER NOT NULL REFERENCES accounts_customuser(user_ptr_id),
    bid_amount_inr NUMERIC(12,2) NOT NULL,
    delivery_timeline_days INTEGER,
    proposal_text TEXT,
    status VARCHAR(50) DEFAULT 'submitted', -- submitted, accepted, rejected
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    updated_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,

    UNIQUE(job_id, contractor_id)
);

-- Milestone Model (Stages of a job)
CREATE TABLE marketplace_milestone (
    id SERIAL PRIMARY KEY,
    job_id INTEGER NOT NULL REFERENCES marketplace_freelancejob(id),
    milestone_number INTEGER NOT NULL,
    title VARCHAR(255) NOT NULL,
    description TEXT,
    deliverable_description TEXT NOT NULL,
    amount_inr NUMERIC(12,2) NOT NULL,
    due_date TIMESTAMP NOT NULL,
    status VARCHAR(50) DEFAULT 'pending', -- pending, in_progress, delivered, approved, paid
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    updated_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,

    INDEX idx_job_id (job_id),
    INDEX idx_status (status)
);

-- Deliverable Model (Work submitted for milestone)
CREATE TABLE marketplace_deliverable (
    id SERIAL PRIMARY KEY,
    milestone_id INTEGER NOT NULL REFERENCES marketplace_milestone(id),
    contractor_id INTEGER NOT NULL REFERENCES accounts_customuser(user_ptr_id),
    submission_text TEXT,
    file_upload_url VARCHAR(500),
    submission_date TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    status VARCHAR(50) DEFAULT 'submitted', -- submitted, approved, revision_requested
    approval_notes TEXT,

    INDEX idx_milestone_id (milestone_id)
);

```

2.4 Payment & Escrow System

```
-- EscrowTransaction Model
CREATE TABLE marketplace_escrowtransaction (
    id SERIAL PRIMARY KEY,
    job_id INTEGER REFERENCES marketplace_freelancejob(id),
    milestone_id INTEGER NOT NULL REFERENCES marketplace_milestone(id),
    amount_inr NUMERIC(12,2) NOT NULL,
    status VARCHAR(50) DEFAULT 'locked', -- locked, released, refunded, disputed
    locked_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    released_at TIMESTAMP,
    refunded_at TIMESTAMP,

    -- Razorpay Integration
    razorpay_order_id VARCHAR(100),
    razorpay_payment_id VARCHAR(100),
    razorpay_signature VARCHAR(255),
    payment_verified BOOLEAN DEFAULT FALSE,

    -- Audit
    transaction_hash UUID UNIQUE NOT NULL, -- Prevent duplicates
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    updated_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,

    INDEX idx_status (status),
    INDEX idx_job_id (job_id),
    INDEX idx_milestone_id (milestone_id)
);

-- WalletTransaction Model
CREATE TABLE marketplace_wallettransaction (
    id SERIAL PRIMARY KEY,
    user_id INTEGER NOT NULL REFERENCES accounts_customuser(user_ptr_id),
    transaction_type VARCHAR(50), -- 'credit', 'debit', 'withdrawal'
    amount_inr NUMERIC(12,2) NOT NULL,
    description VARCHAR(255),
    escrow_transaction_id INTEGER REFERENCES marketplace_escrowtransaction(id),
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,

    INDEX idx_user_id (user_id),
    INDEX idx_created_at (created_at)
);

-- Withdrawal Request Model
CREATE TABLE marketplace_withdrawalrequest (
    id SERIAL PRIMARY KEY,
    user_id INTEGER NOT NULL REFERENCES accounts_customuser(user_ptr_id),
    amount_inr NUMERIC(12,2) NOT NULL,
```

```

bank_account_id INTEGER, -- Reference to encrypted bank details
status VARCHAR(50) DEFAULT 'pending', -- pending, approved, completed, rejected
requested_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
completed_at TIMESTAMP,
transaction_id VARCHAR(100), -- Bank transaction reference
notes TEXT,

INDEX idx_user_id (user_id),
INDEX idx_status (status)
);

```

2.5 Trust & Reputation

-- *TrustSignal Model*

```

CREATE TABLE marketplace_trustsignal (
    id SERIAL PRIMARY KEY,
    user_id INTEGER NOT NULL REFERENCES accounts_customuser(user_ptr_id),
    signal_type VARCHAR(50) NOT NULL, -- 'rating', 'proposal', 'dispute', 'tenure'
    score_delta INTEGER NOT NULL,
    related_object_id INTEGER,
    description VARCHAR(255),
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    expires_at TIMESTAMP, -- For time-decaying signals

    INDEX idx_user_id (user_id),
    INDEX idx_created_at (created_at)
);

```

-- *Dispute Model*

```

CREATE TABLE marketplace_dispute (
    id SERIAL PRIMARY KEY,
    raised_by_id INTEGER NOT NULL REFERENCES accounts_customuser(user_ptr_id),
    job_id INTEGER REFERENCES marketplace_freelancejob(id),
    milestone_id INTEGER REFERENCES marketplace_milestone(id),
    reason VARCHAR(255) NOT NULL,
    description TEXT,
    status VARCHAR(50) DEFAULT 'open', -- open, in_review, resolved, closed
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    resolved_at TIMESTAMP,
    resolution_notes TEXT,
    resolved_by_id INTEGER REFERENCES accounts_customuser(user_ptr_id),

    INDEX idx_status (status),
    INDEX idx_created_at (created_at)
);

```

2.6 Communication

-- ChatThread Model

```
CREATE TABLE marketplace_chatthread (  
    id SERIAL PRIMARY KEY,  
    request_id INTEGER REFERENCES marketplace_helprequest(id),  
    job_id INTEGER REFERENCES marketplace_freelancejob(id),  
    workspace_id INTEGER REFERENCES marketplace_workspace(id),  
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,  
    last_message_at TIMESTAMP,  
    subject VARCHAR(255),  
  
    INDEX idx_request_id (request_id),  
    INDEX idx_job_id (job_id)  
);
```

-- ChatThread Participants (M2M)

```
CREATE TABLE marketplace_chatthread_participants (  
    id SERIAL PRIMARY KEY,  
    chatthread_id INTEGER NOT NULL REFERENCES marketplace_chatthread(id),  
    user_id INTEGER NOT NULL REFERENCES accounts_customuser(user_ptr_id),  
    joined_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,  
  
    UNIQUE(chatthread_id, user_id)  
);
```

-- ChatMessage Model

```
CREATE TABLE marketplace_chatmessage (  
    id SERIAL PRIMARY KEY,  
    thread_id INTEGER NOT NULL REFERENCES marketplace_chatthread(id),  
    sender_id INTEGER NOT NULL REFERENCES accounts_customuser(user_ptr_id),  
    content TEXT NOT NULL,  
    file_attachment_url VARCHAR(500),  
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,  
    updated_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,  
    is_deleted BOOLEAN DEFAULT FALSE,  
    read_by_ids TEXT, -- JSON array of user IDs who read this message  
  
    INDEX idx_thread_id (thread_id),  
    INDEX idx_created_at (created_at)  
);
```

-- Notification Model

```
CREATE TABLE notifications_notification (  
    id SERIAL PRIMARY KEY,  
    user_id INTEGER NOT NULL REFERENCES accounts_customuser(user_ptr_id),  
    notification_type VARCHAR(50), -- 'request_claimed', 'job_proposal', 'milestone_approved'  
    content_type VARCHAR(50), -- 'help_request', 'freelance_job', etc.
```

```

object_id INTEGER,
title VARCHAR(255),
message TEXT,
is_read BOOLEAN DEFAULT FALSE,
created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,

INDEX idx_user_id (user_id),
INDEX idx_is_read (is_read),
INDEX idx_created_at (created_at)
);

```

2.7 Workspace & Collaboration

-- Workspace Model

```

CREATE TABLE marketplace_workspace (
    id SERIAL PRIMARY KEY,
    name VARCHAR(255) NOT NULL,
    owner_id INTEGER NOT NULL REFERENCES accounts_customuser(user_ptr_id),
    description TEXT,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    updated_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,

    INDEX idx_owner_id (owner_id)
);

```

-- Workspace Members (M2M)

```

CREATE TABLE marketplace_workspace_members (
    id SERIAL PRIMARY KEY,
    workspace_id INTEGER NOT NULL REFERENCES marketplace_workspace(id),
    user_id INTEGER NOT NULL REFERENCES accounts_customuser(user_ptr_id),
    role VARCHAR(50) DEFAULT 'developer', -- owner, maintainer, developer
    joined_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,

    UNIQUE(workspace_id, user_id)
);

```

-- Sprint Model

```

CREATE TABLE marketplace_sprint (
    id SERIAL PRIMARY KEY,
    workspace_id INTEGER NOT NULL REFERENCES marketplace_workspace(id),
    name VARCHAR(255) NOT NULL,
    start_date TIMESTAMP NOT NULL,
    end_date TIMESTAMP NOT NULL,
    status VARCHAR(50) DEFAULT 'open', -- open, in_progress, completed
    goal TEXT,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,

    INDEX idx_workspace_id (workspace_id)
);

```

```

);

-- Issue/Task Model
CREATE TABLE marketplace_issue (
    id SERIAL PRIMARY KEY,
    workspace_id INTEGER NOT NULL REFERENCES marketplace_workspace(id),
    sprint_id INTEGER REFERENCES marketplace_sprint(id),
    title VARCHAR(255) NOT NULL,
    description TEXT,
    assigned_to_id INTEGER REFERENCES accounts_customuser(user_ptr_id),
    status VARCHAR(50) DEFAULT 'todo', -- todo, in_progress, done
    priority VARCHAR(20) DEFAULT 'medium',
    created_by_id INTEGER NOT NULL REFERENCES accounts_customuser(user_ptr_id),
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,

    INDEX idx_workspace_id (workspace_id),
    INDEX idx_sprint_id (sprint_id),
    INDEX idx_status (status)
);

```

2.8 AI & Moderation

```

-- AIAssistanceLog Model
CREATE TABLE marketplace_aiassistancelog (
    id SERIAL PRIMARY KEY,
    user_id INTEGER NOT NULL REFERENCES accounts_customuser(user_ptr_id),
    request_id INTEGER REFERENCES marketplace_helprequest(id),
    assistance_type VARCHAR(50), -- 'summary', 'matching', 'moderation'
    input_tokens INTEGER,
    output_tokens INTEGER,
    cost_usd NUMERIC(10,4),
    response_json TEXT,
    status VARCHAR(50), -- 'success', 'failed'
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,

    INDEX idx_user_id (user_id),
    INDEX idx_created_at (created_at)
);

```

```

-- ContentFlag Model (Moderation)
CREATE TABLE marketplace_contentflag (
    id SERIAL PRIMARY KEY,
    reported_by_id INTEGER NOT NULL REFERENCES accounts_customuser(user_ptr_id),
    content_type VARCHAR(50), -- 'help_request', 'freelance_job', 'comment'
    object_id INTEGER NOT NULL,
    reason VARCHAR(50), -- 'spam', 'hate_speech', 'scam'
    description TEXT,
    status VARCHAR(50) DEFAULT 'open', -- open, in_review, resolved

```

```

    moderator_notes TEXT,
    action_taken VARCHAR(50), -- 'none', 'warning', 'content_removed', 'user_suspended'
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    resolved_at TIMESTAMP,

    INDEX idx_status (status),
    INDEX idx_created_at (created_at)
);

```

3. Key Indexes for Performance

-- User Queries

```

CREATE INDEX idx_customuser_email ON accounts_customuser(email);
CREATE INDEX idx_customuser_trust_score ON accounts_customuser(trust_score DESC);
CREATE INDEX idx_customuser_created_at ON accounts_customuser(created_at);

```

-- HelpRequest Queries

```

CREATE INDEX idx_helprequest_status_created ON marketplace_helprequest(status, created_at DESC);
CREATE INDEX idx_helprequest_claimed_by_status ON marketplace_helprequest(claimed_by_id, status);
CREATE INDEX idx_helprequest_requester_status ON marketplace_helprequest(requester_id, status);

```

-- FreelanceJob Queries

```

CREATE INDEX idx_freelancejob_status_deadline ON marketplace_freelancejob(status, deadline);
CREATE INDEX idx_freelancejob_client_status ON marketplace_freelancejob(client_id, status);
CREATE INDEX idx_freelancejob_contractor ON marketplace_freelancejob(contractor_id);

```

-- Payment Queries

```

CREATE INDEX idx_escrow_job_status ON marketplace_escrowtransaction(job_id, status);
CREATE INDEX idx_wallet_transaction_user_date ON marketplace_wallettransaction(user_id, created_at);

```

-- Communication Queries

```

CREATE INDEX idx_chatmessage_thread_date ON marketplace_chatmessage(thread_id, created_at DESC);
CREATE INDEX idx_notification_user_read_date ON notifications_notification(user_id, is_read, created_at);

```

-- Search (Full-text)

```

CREATE INDEX idx_helprequest_search_vector ON marketplace_helprequest USING GIN(search_vector);
CREATE INDEX idx_freelancejob_search_vector ON marketplace_freelancejob USING GIN(search_vector);

```

4. Database Constraints

-- Foreign Key Constraints

```

ALTER TABLE marketplace_helprequest ADD CONSTRAINT fk_requester
    FOREIGN KEY (requester_id) REFERENCES accounts_customuser(user_ptr_id);

```

```

ALTER TABLE marketplace_helprequest ADD CONSTRAINT fk_claimed_by

```

```

    FOREIGN KEY (claimed_by_id) REFERENCES accounts_customuser(user_ptr_id);

-- Check Constraints
ALTER TABLE accounts_customuser ADD CONSTRAINT check_kp_non_negative
    CHECK (knowledge_points >= 0);

ALTER TABLE accounts_customuser ADD CONSTRAINT check_wallet_non_negative
    CHECK (wallet_inr >= 0.00);

ALTER TABLE marketplace_helprequest ADD CONSTRAINT check_kp_bounty_range
    CHECK (kp_bounty BETWEEN 1 AND 1000);

ALTER TABLE marketplace_rating ADD CONSTRAINT check_rating_range
    CHECK (score BETWEEN 1 AND 5);

-- Unique Constraints
ALTER TABLE accounts_customuser ADD CONSTRAINT unique_username UNIQUE (username);
ALTER TABLE accounts_customuser ADD CONSTRAINT unique_email UNIQUE (email);
ALTER TABLE marketplace_escrowtransaction ADD CONSTRAINT unique_transaction_hash UNIQUE (trans

```

5. Normalization & Query Optimization

Normalization Level: 3NF (Third Normal Form) - No transitive dependencies - All non-key attributes depend on primary key - No partial dependencies

Denormalization Strategies (for performance): - CustomUser.trust_score: Cached, updated via signals - HelpRequest.ai_summary: Pre-computed once - CustomUser.knowledge_points: Real-time but critical, frequent reads - Notification.is_read: Frequently queried, indexed

Summary

The HelperLearner database: - Normalized to 3NF - 20+ tables covering all business logic - Proper foreign key relationships - Comprehensive indexes for common queries - ACID properties via PostgreSQL - Audit columns (created_at, updated_at) - Supports horizontal scaling (via sharding on user_id if needed)